Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines admit classical estimation of training observables for suitable input states, allowing model optimization without quantum circuit sampling. However, their standard fixed-input formulation models a single distribution. In this work, we introduce a conditional framework that encodes conditions into input states before applying a learned IQP unitary shared across conditions. The input preparation need not have strict IQP form, provided its reference state supports efficient classical sampling and amplitude-ratio evaluation. These operations enable unbiased classical estimation of Pauli-Z moments and their gradients. Experiments on sixteen-qubit discrete Gaussian families and the Curie–Weiss model demonstrate conditional generation. A correlated Gaussian encoder supports two-dimensional conditioning and generalization to mean–width combinations absent from fitting and development. For Curie–Weiss, the shared model achieves lower mean full-distribution total variation distance than both independent baseline families at two of three held-out temperatures, although the baselines are fitted directly at those temperatures and each family uses three times the total parameters and optimizer updates. Additional runs demonstrate the computational feasibility of classical conditional training from 64 to 2048 qubits. A sixteen-qubit signed-graph construction further combines correlated temperature encoding with exact switching symmetry and finite-data learning of two shared angles.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a model $q_{\boldsymbol{\theta}}(\cdot \mid c)$, where c denotes the condition and $\boldsymbol{\theta}$ contains learned parameters shared across conditions. Classical examples include conditional generative adversarial networks [1–3], conditional variational autoencoders [4–6], and conditional diffusion models [7–9]. In the quantum domain, researchers have also adapted quantum circuit Born machines (QCBMs) [10, 11] to conditional tasks [12]. Researchers can introduce conditioning through input-state preparation or dedicated condition registers before applying a shared parameterized circuit [13–16]. Alternatively, postselection on a learned joint Born distribution can also yield conditional samples [17]. Beyond QCBMs, quantum diffusion models have explored conditional generation of both classical data and quantum states [18–20].

Existing complexity-theoretic results support the conjecture that classical algorithms cannot efficiently sample from certain output families of instantaneous quantum polynomial-time (IQP) Born machines under standard assumptions [21–25]. Yet, for maximum mean discrepancy (MMD) training with a Gaussian kernel [26], as commonly used for QCBMs [10, 27], classical algorithms can efficiently estimate the required Pauli-Z expectation values and their parameter gradients [28]. These estimators build on Van den Nest's techniques for computationally tractable input states [29]. However, the standard IQP formulation models a single target distribution per circuit. Extending IQP Born machines to conditional generation requires a conditioning mechanism that preserves classical access to the training observables and their gradients.

In this work, we develop a classically trainable framework for conditional IQP Born machines. We apply one learned IQP unitary $U_{\boldsymbol{\theta}}$ to condition-dependent input states $|\phi(c)\rangle$, giving $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. We establish sufficient conditions on the encoder for efficient classical estimation of the conditional training observables and their gradients, and construct encoders satisfying these conditions. Specifically, for the reference amplitudes $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$, we require efficient sampling from $|\psi_c|^2$ and evaluation of amplitude ratios.

Building on sparse-observable methods [29, 30] and IQP training methods [28], we derive unbiased estimators of conditional Pauli-Z moments and their derivatives with respect to the learned phases. We bound the second moments of the moment and derivative estimators by one and four, respectively. These bounds imply that $O(\epsilon^{-2})$ reference samples suffice to estimate each specified moment or derivative to additive error ϵ at fixed confidence. We combine these estimators to obtain unbiased finite-sample estimates of the conditional MMD^2 objective and its gradients, enabling classical optimization of a single IQP Born machine across conditions. Figure 1 summarizes the framework.

We evaluate the framework on discrete Gaussian families and Curie–Weiss distributions at sixteen qubits. A

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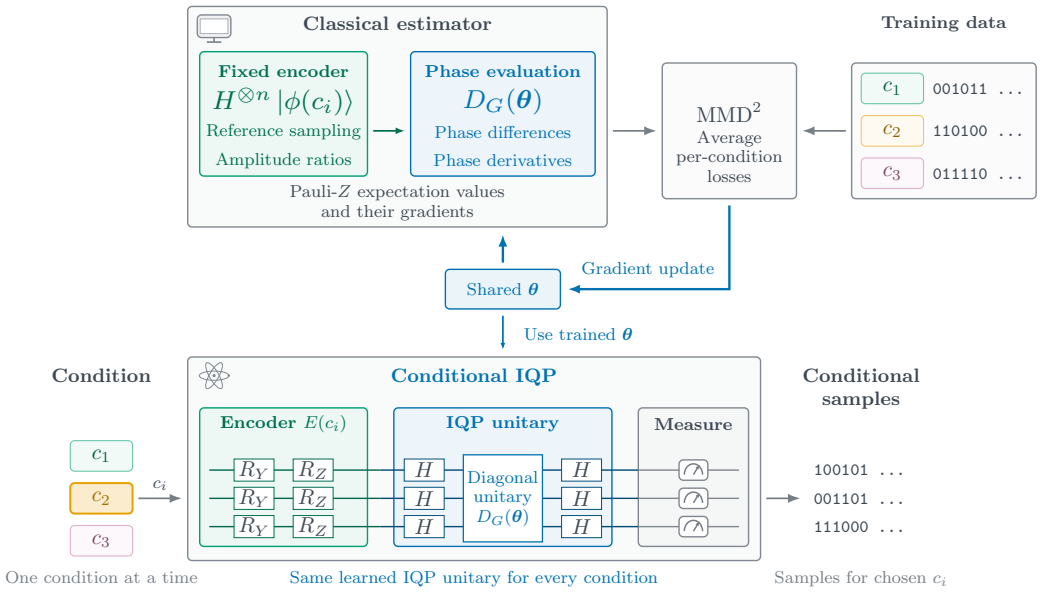


FIG. 1. Proposed framework for classical training and quantum sampling of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-Z moments and their gradients. We use these estimates to optimize one shared parameter vector $\boldsymbol{\theta}$ with a squared maximum mean discrepancy (MMD) objective averaged across training conditions. During quantum sampling, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

product-state encoder first illustrates conditional translation of Gaussian profiles. We then encode both mean and width in a correlated input and train a shared model with sixteen local rotations. Training the two coordinates separately tests generalization to unseen joint combinations, including pairs outside the joint training hull but within both marginal ranges.

For the Curie–Weiss model, we train a single model at four inverse temperatures. The model captures the change from unimodal to bimodal magnetization profiles, including the emergence of bimodality at the held-out inverse temperature $\beta = 1.0$. The encoder represents temperature through input phases while maintaining uniform computational-basis probabilities. The learned circuit converts this phase dependence into temperature-dependent output distributions. We compare this model with conditional and unconditional baseline families. Each family contains three independent circuits that we fit directly at the evaluation temperatures and uses three times the total parameters and optimizer updates of the shared model. The shared model achieves lower mean full-distribution total variation distance (TVD) than both baseline families at two of the three held-out temperatures, including $\beta = 1.6$, beyond the training range.

We also assess computational feasibility through optimization runs from 64 to 2048 qubits, tracking sampled MMD^2 at held-out temperatures throughout training.

Input-state design also allows exact output symmetries without restricting the learned IQP phases. A sixteen-qubit signed-graph experiment encodes temperature through correlated triangle phases and imposes sixty-four vertex switches at every condition. We train two shared local angles using 128 target samples at each of four temperatures and evaluate conditional graph order beyond the fitting range. Appendix C specifies the reference access, finite-data protocol and accuracy limitations.

The remainder of this paper is organized as follows. In Section II, we introduce the conditional IQP model and its classical estimators. In Section III, we develop the conditional MMD training protocol. In Section IV, we describe the numerical experiments and report their results. In Section V, we show how input-state preparation can enforce output symmetries. In Section VI, we discuss the implications and limitations of our findings, and in Section VII, we summarize the contributions.

II. CONDITIONAL IQP MODEL

We separate condition encoding from the learned unitary transformation. An encoder maps each condition $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, and we apply the same parameterized IQP unitary $U_{\boldsymbol{\theta}}$ to every encoded state. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings,

we define the model distribution as

$$q_{\boldsymbol{\theta}}(x \mid c) = \left|\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle\right|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

Throughout this work, we use "IQP unitary" to denote only the learned block $U_{\boldsymbol{\theta}}$. The encoder need not have IQP form and may include non-diagonal operations, so the complete conditional circuit need not belong to the strict IQP circuit class. Our conditions constrain classical access to the reference state rather than the encoder's gate structure: the reference state must support efficient sampling and amplitude-ratio evaluation.

The encoder also constrains how the model's output probabilities depend on the condition c. For rotation encodings with angles linear in c, the encoding generators determine the accessible Fourier frequencies, while the learned parameters control their coefficients within the constraints of the circuit architecture [31]. Multidimensional Fourier analyses further describe how the encoding and circuit architecture constrain the model's expressive capacity [32]. Caro et al. also derive encoding-dependent generalization bounds for parameterized quantum circuits [33]. We therefore distinguish encoder expressivity and generalization from the requirements for efficient classical estimation.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. We write the unitary as

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j\right). \tag{3}$$

All generators in D_G are diagonal in the computational basis and commute. In the numerical experiments, we use only one- and two-body generators. However, this formulation extends naturally to k-body interactions by adding the corresponding Pauli-Z products to the exponent.

We train the model using samples and amplitude ratios from the reference state $|\psi_c\rangle$ immediately before the diagonal IQP layer, without sampling the circuit's output distribution. We define its computational-basis amplitudes as

$$\psi_c(z) = \langle z| H^{\otimes n} |\phi(c)\rangle, \tag{4}$$

The squared amplitudes $|\psi_c(z)|^2$ give the reference sampling distribution for condition c. We require encoders that support two efficient classical operations for the relevant conditions c and bit masks a:

1. Sample $z \sim |\psi_c(z)|^2$.

2. Evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled z, where $\oplus$ denotes bitwise addition modulo two.

We call these two operations the sampling-and-ratio interface.

We estimate Pauli-Z moments by combining reference amplitude ratios with phase factors from the IQP layer. For each sampled string z, the ratio $R_c(z, a)$ compares the reference amplitudes of z and the paired string $z \oplus a$. The IQP layer contributes a complex phase factor determined by the phase difference between these strings. We multiply the amplitude ratio by this phase factor and average the real part over reference samples to estimate the corresponding Pauli-Z moment [29].

We evaluate amplitude ratios only on the support of $|\psi_c|^2$, because reference sampling never selects strings with $\psi_c(z) = 0$. Computationally tractable reference states in the sense of Van den Nest admit efficient computational-basis sampling and amplitude evaluation [29]. These states provide a sufficient realization of the sampling-and-ratio interface when amplitude evaluation supplies the precision needed to compute the required ratios efficiently.

The simplest case is a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^n |\phi_i(c)\rangle$. Its reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b| H |\phi_i(c)\rangle$. We can therefore generate reference samples by drawing each bit independently, and the amplitude ratio is

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Existing IQP sampling-hardness results apply to specific circuit families and do not establish hardness for arbitrary condition-dependent input states or trained parameter values [22, 23]. To generate conditional samples on a quantum computer, we also need efficient circuits that prepare $|\phi(c)\rangle$ and implement $U_{\boldsymbol{\theta}}$. The classical sampling-and-ratio interface alone does not guarantee such circuits.

A. Conditional Pauli estimator and gradients

We express the conditional MMD objective in Section III in terms of Pauli-Z moments. For a bit mask $a \in \{0,1\}^n$, we define $Z_a = \bigotimes_{i=1}^n Z_i^{a_i}$ and denote its conditional model expectation by $\mu_q(a \mid c)$. We use the sampling-and-ratio interface to derive estimators for this

moment and its derivatives with respect to the learned phases.

We denote the diagonal IQP layer by $D_{\boldsymbol{\theta}}$ and write its action as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle , \tag{7}$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

where P is the number of generators and $g_j \in \{0,1\}^n$ specifies the support of the jth Pauli-Z generator. We evaluate the dot product $g_j \cdot z$ modulo two. We define the phase difference associated with the mask a as

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{9}$$

For a reference sample $z \sim |\psi_c|^2$, we define

$$\begin{aligned} W_{c,a}(z;\boldsymbol{\theta}) &= R_c(z,a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z;\boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}). \end{aligned} \tag{10}$$

We define the derivative of the phase difference with respect to θ_j as

$$\begin{aligned} d_{j,a}(z) &= \partial_{\theta_j} \Delta_a f_{\boldsymbol{\theta}}(z) \\ &= \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \end{aligned} \tag{11}$$

The following proposition establishes unbiased estimators for conditional Pauli-Z moments and their phase derivatives, and bounds their sampling error. We obtain these estimators by adapting amplitude-ratio methods for sparse observables [30]. For computationally tractable reference states, efficient estimation also follows from Van den Nest's sparse-operator theorem [29].

Proposition 1 (Classical estimation from reference-state access)**.** *Fix a condition c, a bit mask a, and a phase function with polynomially many diagonal Pauli-Z terms, as in Eq. (8). We assume that the reference state is normalized and independent of $\boldsymbol{\theta}$. We also assume that we can efficiently draw independent samples from $|\psi_c|^2$ and evaluate $R_c(z,a)$ for each sampled z. For a reference sample $z \sim |\psi_c|^2$, we define the moment and derivative estimators as follows:*

$$\begin{aligned} Y &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}), \\ Y'_j &= \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})] \end{aligned} \tag{12}$$

With exact sampling and arithmetic, these estimators have the following expectations and second-moment bounds:

$$\begin{aligned} \mathbb{E}[Y] &= \mu_q(a \mid c), & \mathbb{E}[Y^2] &\le 1, \\ \mathbb{E}[Y'_j] &= \partial_{\theta_j} \mu_q(a \mid c), & \mathbb{E}[(Y'_j)^2] &\le 4. \end{aligned} \tag{13}$$

We obtain unbiased estimates of the moment and its phase derivative by averaging Y and Y'_j over S independent reference samples. Their mean-square errors are at most $1/S$ and $4/S$, respectively. We can therefore estimate a specified moment or phase derivative to additive error ϵ at fixed confidence using $S = O(\epsilon^{-2})$ reference samples. These guarantees hold even when some reference amplitudes vanish.

Proof. We define the support of the reference distribution as $S_c = \{z : \psi_c(z) \neq 0\}$. Using $HZH = X$, we conjugate the measured observable Z_a through the final Hadamard layer to obtain $X_a = \bigotimes_{i=1}^n X_i^{a_i}$. This operator flips the bits selected by a, so

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}} |\psi_c\rangle \\ &= \mathrm{Re} \sum_{z \in S_c} \psi_c(z)^* \psi_c(z \oplus a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)} \\ &= \mathbb{E}_{z \sim |\psi_c|^2}[Y]. \end{aligned} \tag{14}$$

The last equality rewrites the sum as an expectation under $|\psi_c|^2$, using the amplitude ratio and the definition of Y.

The reference state does not depend on $\boldsymbol{\theta}$, so its sampling distribution and amplitude ratios remain fixed. We therefore differentiate only the IQP phase factor in the finite sum to obtain

$$\begin{aligned} \partial_{\theta_j} \mu_q(a \mid c) &= \mathbb{E}_{z \sim |\psi_c|^2}[\mathrm{Re}\,(i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta}))] \\ &= \mathbb{E}[Y'_j]. \end{aligned} \tag{15}$$

Thus Y'_j is an unbiased estimator of the phase derivative.

Although individual amplitude ratios can be large, normalization bounds their second moment. The phase factor has unit magnitude, and the sampling probability $|\psi_c(z)|^2$ cancels the denominator in $|R_c(z,a)|^2$ when we take the expectation. We therefore obtain

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z \in S_c} |\psi_c(z \oplus a)|^2 \le 1. \tag{16}$$

The final inequality follows because bit flipping maps S_c to a subset of all bit strings and the reference state is normalized. Since Y is the real part of $W_{c,a}$, we have $\mathbb{E}[Y^2] \le 1$. Combining $|d_{j,a}(z)| \le 2$ with the same bound gives $\mathbb{E}[(Y'_j)^2] \le 4$. Averaging Y and Y'_j over S independent reference samples therefore gives mean-square errors at most $1/S$ and $4/S$, respectively. Chebyshev's inequality then yields the stated fixed-confidence sample complexity. The argument allows vanishing reference amplitudes because we evaluate ratios only for sampled strings. □

In practice, we draw S independent reference samples $z_s \sim |\psi_c|^2$ and estimate the moment as

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{17}$$

We estimate the phase derivative by averaging the corresponding Y'_j values over the same samples. For the

standard IQP input $|0\rangle^{\otimes n}$, all reference amplitudes are equal, so $R_c(z,a)=1$. The moment estimator then reduces to $Y=\cos[\Delta_a f_{\boldsymbol{\theta}}(z)]$, recovering the usual IQP cosine estimator [29, 34].

For a circuit with r ancillary qubits, we apply the same estimator to the joint reference state, provided it supports the sampling-and-ratio interface. If we discard the ancillary measurement outcomes, we extend each visible-bit mask a to $(a,0^r)$, where 0^r denotes r zeros. The zero entries ensure that the observable acts as the identity on the ancillary qubits.

III. CONDITIONAL IQP TRAINING

We now formulate the training protocol for conditional IQP Born machines. Throughout this work, we use *classically trainable* to mean that we can efficiently estimate the training objective and its phase gradients to a prescribed additive accuracy on a classical computer.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population squared MMD is [26]

$$\begin{aligned}\mathrm{MMD}^2(p,q)=\mathbb{E}_{x,x'\sim p}[k(x,x')]-2\mathbb{E}_{x\sim p,\,y\sim q}[k(x,y)]\\+\mathbb{E}_{y,y'\sim q}[k(y,y')],\end{aligned}\tag{18}$$

where we draw the two samples in each expectation independently.

Throughout this work, we use the Gaussian kernel on binary strings,

$$k_\sigma(x,y)=\exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right],\tag{19}$$

where $d_H(x,y)$ is the Hamming distance, which counts the positions at which x and y differ, and $\sigma>0$ is the kernel bandwidth.

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We train the shared model by minimizing the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta})=\frac{1}{|\mathcal{C}_{\mathrm{train}}|}\sum_{c\in\mathcal{C}_{\mathrm{train}}}\frac{1}{|\tau|}\sum_{\sigma\in\tau}\mathrm{MMD}^2_\sigma(p(\cdot\mid c),q_{\boldsymbol{\theta}}(\cdot\mid c)).\tag{20}$$

Here, MMD^2_σ denotes the squared MMD computed with the kernel k_σ. We weight all training conditions and bandwidths equally. At each optimizer update, we independently draw one condition uniformly from $\mathcal{C}_{\mathrm{train}}$ and one bandwidth uniformly from τ, giving a stochastic estimate of this average.

B. Character expansion

For binary strings, the Gaussian kernel admits an exact expansion in binary characters. We sample a visible-bit mask $a\in\{0,1\}^n$ by drawing its coordinates independently according to

$$a_i\sim\mathrm{Bernoulli}(p_\sigma),\qquad p_\sigma=\frac{1-e^{-1/(2\sigma^2)}}{2},\tag{21}$$

The corresponding binary character is

$$\chi_a(x)=(-1)^{a\cdot x},\tag{22}$$

which records the parity of the bits selected by a. Averaging the product of these characters over a gives

$$k_\sigma(x,y)=\mathbb{E}_a[\chi_a(x)\chi_a(y)].\tag{23}$$

For a condition c, define the target and model character moments

$$\begin{aligned}\mu_p(a\mid c)&=\mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)],\\ \mu_q(a\mid c)&=\mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned}\tag{24}$$

Substituting Eq. (23) into Eq. (18) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot\mid c),q_{\boldsymbol{\theta}}(\cdot\mid c))=\mathbb{E}_a\left[(\mu_p(a\mid c)-\mu_q(a\mid c))^2\right].\tag{25}$$

Thus, the squared MMD averages the squared differences between target and model character moments over the mask distribution.

For a Born machine, the visible Pauli operator

$$Z_a=\bigotimes_{i=1}^n Z_i^{a_i}\tag{26}$$

has eigenvalue $\chi_a(x)$ on the computational-basis state $|x\rangle$. Its model expectation value is therefore

$$\mu_q(a\mid c)=\langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}.\tag{27}$$

We can thus express the conditional squared MMD entirely in terms of visible Pauli-Z moments. We estimate the model moments classically using the reference-state estimator in Sec. II A.

The bandwidth σ determines which correlation orders the objective emphasizes. The mask's Hamming weight $K=|a|$ counts the bits it selects, so it gives the correlation order of Z_a. Because we sample the mask coordinates independently,

$$\begin{aligned}K&\sim\mathrm{Binomial}(n,p_\sigma),\\ \mathbb{E}[K]&=np_\sigma,\\ \mathrm{Var}(K)&=np_\sigma(1-p_\sigma).\end{aligned}\tag{28}$$

Each bandwidth therefore induces a distribution over correlation orders. Large bandwidths favor lower-order

correlations, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The choice of bandwidth can also influence the model's trainability. Previous results demonstrate that cost functions consisting of global terms can induce barren plateaus [35, 36], even in shallow quantum circuits [37]. This creates a trade-off between trainability and sensitivity to higher-order correlations. Improving trainability may require restricting the objective to low-body correlations, which limits its ability to distinguish distributions that differ in higher-order correlations [38]. Researchers have proposed several strategies to address these difficulties, including tailored generator sets and restricted initializations [39, 40], and adversarial kernel adaptation [41]. How best to balance trainability and sensitivity to higher-order correlations remains an open question beyond the scope of this work.

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and visible mask a, we draw $B \geq 2$ independent target samples and compute

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c). \tag{29}$$

Independently of these target samples, we draw $S \geq 2$ independent reference samples and evaluate the reference-state estimator from Eq. (10),

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2. \tag{30}$$

We denote the corresponding sample means by

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{31}$$

The direct estimate $(\overline{Y} - \overline{T})^2$ overestimates the squared moment difference on average because it includes the sampling variances of both sample means. We remove this bias by excluding products that use the same draw twice:

$$\begin{aligned}\widehat{\Delta_c^2}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}.\end{aligned} \tag{32}$$

Independence within each pair gives

$$\mathbb{E}\left[\widehat{\Delta_c^2}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{33}$$

Thus, Eq. (32) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (25). An unbiased finite-sample estimate of MMD^2 may be negative.

We evaluate the pair sums using the identities

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{34}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{35}$$

Once we have computed the sample values, we can therefore evaluate the estimator in time linear in the numbers of target and reference samples.

The same construction gives an unbiased estimator of the phase gradients. For a phase parameter θ_j, let

$$Y_s' = \partial_{\theta_j} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{36}$$

We evaluate Y_s' using Eq. (15). Differentiating Eq. (32) gives

$$\partial_{\theta_j}\widehat{\Delta_c^2}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y_t' - 2\overline{T}\,\overline{Y'}, \tag{37}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_s Y_s'. \tag{38}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{39}$$

which is the gradient of the corresponding squared moment difference.

At a fixed parameter vector, the expectations above average over independent draws from the target distribution and the reference sampling law. During training, we sample target minibatches with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the empirical objective associated with those pools.

We exclude the diagonal terms $s = t$ because Y_s and Y_s' can be correlated when we compute them from the same reference draw. The fixed encoder makes the reference sampling law independent of $\boldsymbol{\theta}$. We therefore hold the sampled reference strings fixed during automatic differentiation to evaluate the phase gradients without differentiating through the sampler.

D. Stochastic mask estimator

Equation (25) expresses the squared MMD as an expectation over the mask distribution in Eq. (21). Since there are 2^n possible masks, we estimate this expectation using a finite sample of masks.

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{40}$$

for every condition and therefore contributes zero to the squared MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{41}$$

Under the original mask distribution, nonidentity masks have total probability

$$Q_\sigma^+ = 1 - (1 - p_\sigma)^n. \tag{42}$$

We multiply the conditional expectation by Q_σ^+ to preserve the original Gaussian kernel objective.

We draw M nonidentity masks $a_1, \ldots, a_M$ independently from $P_\sigma(a \mid a \neq 0)$ and estimate the squared MMD at condition c as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta}_c^2(a_m), \tag{43}$$

where $\widehat{\Delta}_c^2$ is the unbiased finite-sample estimator in Eq. (32). Averaging over the mask, target, and reference draws recovers the full kernel squared MMD.

We evaluate each distinct sampled mask only once. Let $\mathcal{U}$ denote the set of distinct masks and m_u the number of times mask u occurs. We rewrite Eq. (43) as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \sum_{u \in \mathcal{U}} \frac{m_u}{M} \widehat{\Delta}_c^2(u). \tag{44}$$

Weighting each distinct mask by m_u/M preserves the sampled objective and its gradient while reducing the number of Pauli moments we evaluate.

IV. NUMERICAL EXPERIMENTS

We evaluate the proposed architecture on Gaussian families controlled by translation or by both mean and width, and on the Curie–Weiss model at sixteen qubits. We also assess computational feasibility through optimization runs from 64 to 2048 qubits. For the sixteen-qubit benchmarks, we use the exact full-distribution total variation distance as the primary accuracy metric,

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_{x \in \{0,1\}^n} |p(x) - q(x)|. \tag{45}$$

We also report the exact TVD between Hamming-weight profiles as a diagnostic,

$$\mathrm{TVD}_K(p, q) = \frac{1}{2} \sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{46}$$

where $\rho_p(k)$ and $\rho_q(k)$ give the probabilities of observing Hamming weight k under p and q, respectively. Grouping bitstrings by Hamming weight gives $\mathrm{TVD}_K(p, q) \leq \mathrm{TVD}(p, q)$. The targets assign equal probability to all bitstrings with the same Hamming weight, but the learned models need not preserve this property. Profile TVD therefore does not resolve errors within Hamming-weight shells. We compute both metrics using the same exact circuit probabilities.

We train the models by minimizing the squared-MMD objective. We select checkpoints for the one-dimensional Gaussian translation experiment using mean exact profile TVD over the development conditions. For the two-dimensional Gaussian and all trained sixteen-qubit Curie–Weiss models, we use exact full-distribution TVD for hyperparameter and checkpoint selection. Appendices A and B specify the training and evaluation protocols.

A. Discrete Gaussian translation

Our first experiment illustrates the conditioning mechanism with a translated discrete Gaussian Hamming-weight profile. For $n = 16$ bits, let $K = |x|$ denote the Hamming weight of bitstring x. The condition $c \in [0, 1]$ controls the nominal center $5 + 6c$ of the profile. The target distribution assigns equal probability to all bitstrings with the same Hamming weight.

We use a fixed product-state encoder that applies the same two rotations to every qubit:

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z\left(c - \frac{1}{2}\right) R_Y(1.5)\, |0\rangle_i \,. \tag{47}$$

Changing c modifies the input phases but leaves the encoder's computational-basis measurement distribution unchanged.

The shared unitary uses a 4×4 lattice with 16 one-body phase gates and 24 two-body phase gates between horizontal and vertical nearest neighbors, giving 40 trainable parameters.

We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$ and evaluate at 80 held-out conditions. The median across runs of the mean full-distribution TVD over these conditions is 0.0995, compared with 0.2922 for the encoder alone (Table I). The corresponding profile TVD is 0.0464. The higher full-distribution TVD reveals discrepancies within Hamming-weight shells that the profile metric does not resolve. Figure 2 shows the translated profiles and conditional response. Appendix A 2 gives the training and evaluation protocol.

B. Two-dimensional Gaussian conditioning and generalization

We next extend the Gaussian benchmark to two conditioning variables that control the profile center and

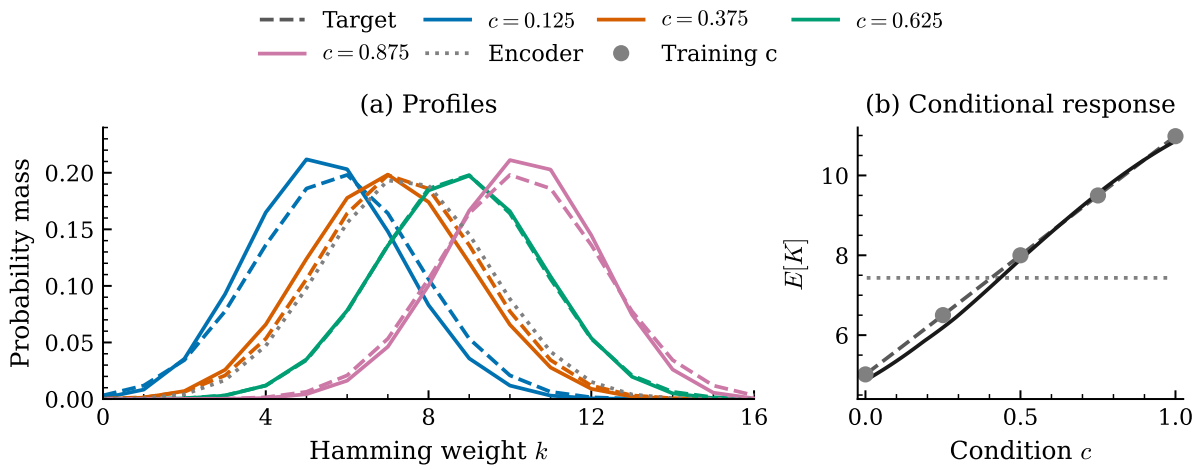


FIG. 2. Conditional translation of discrete Gaussian Hamming-weight profiles at sixteen qubits. (a) Solid colored curves show exact model profiles averaged across five runs at the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. We use these conditions for checkpoint selection and exclude them from fitting. (b) The solid black curve shows the median across runs of the model's mean output Hamming weight at each of 80 held-out conditions. Shading shows the interquartile range. Dashed curves show exact targets, dotted gray curves show the encoder alone, and gray circles mark training conditions on the target response. Table I reports exact full-distribution and profile errors.

TABLE I. Exact full-distribution and profile TVDs for the Gaussian benchmark. Each entry reports the median across five runs of the mean error over 80 held-out conditions. Bold entries mark the lowest error in each column.

Model	Full TVD ↓	Profile TVD ↓
Conditional IQP	**0.0995**	**0.0464**
Encoder only	0.2922	0.2922

width. For sixteen bits and Hamming weight $K = |x|$, we define the target distribution as

$$p(x \mid u, v) = \frac{\rho_{u,v}(K)}{\binom{16}{K}}, \tag{48}$$

$$\rho_{u,v}(k) \propto \exp\left[-\frac{(k-\mu(u))^2}{2\sigma_{\mathrm{G}}(v)^2}\right], \tag{49}$$

where $u, v \in [-1, 1]$, $\mu(u) = 8 + 3u$, and $\sigma_{\mathrm{G}}(v) = 2(1 + 0.2v)$. The coordinate u controls the profile center over $[5, 11]$, while v controls its width over $[1.6, 2.4]$. We normalize the profile over $k \in \{0, \ldots, 16\}$ and assign equal probability to all bitstrings with the same Hamming weight. Because we discretize and truncate the Gaussian profile, $\mu(u)$ and $\sigma_{\mathrm{G}}(v)$ need not equal its actual mean and standard deviation.

We construct the conditional input state in two steps. First, we encode the width coordinate v through a Gaussian Hamming-weight profile centered at $k = 8$:

$$r_v(k) \propto \exp\left[-\frac{(k-8)^2}{2\tau(v)^2}\right], \qquad \tau(v) = \frac{2}{1 + 0.2v}. \tag{50}$$

We normalize this profile over $k \in \{0, \ldots, 16\}$. Increasing v narrows the input profile, whereas it widens the target profile.

We then use the square roots of these probabilities as amplitudes:

$$|\chi(v)\rangle = \sum_{k=0}^{16} \sqrt{r_v(k)}\, |D_k^{(16)}\rangle\,. \tag{51}$$

Here, $|D_k^{(16)}\rangle$ denotes the normalized uniform superposition of all bitstrings with Hamming weight k. Measuring $|\chi(v)\rangle$ therefore produces the profile $r_v(k)$, with equal probabilities for bitstrings of the same weight.

Second, we encode the center coordinate u by applying the same phase rotation $R_Z(u/2)$ to every qubit. Combining both steps gives

$$\begin{aligned} |\phi(u, v)\rangle &= R_Z(u/2)^{\otimes 16}\, |\chi(v)\rangle \\ &= \sum_{k=0}^{16} \sqrt{r_v(k)}\, e^{iu(k-8)/2}\, |D_k^{(16)}\rangle\,. \end{aligned} \tag{52}$$

Thus, v controls the input probabilities, while u controls the relative phases between Hamming-weight sectors.

This encoder supplies a task-specific Gaussian prior, and the learned circuit transforms these inputs to approximate the target family. The input measurement profile remains centered at $k = 8$ for every u, and its width varies oppositely to the target width as v increases. Except at $(u, v) = (0, 0)$, the input and target distributions therefore differ. The circuit must convert the encoded phases into changes in the output center and transform the input width response to approximate the target family. We fix the encoder throughout training and use its

measurement distribution as a baseline to assess the improvement achieved by the learned circuit.

We apply the same learned layer to every encoded input:

$$U_{\boldsymbol{\theta}} = \exp\left(i\sum_{j=1}^{16}\theta_j X_j\right) = \bigotimes_{j=1}^{16} R_X(-2\theta_j). \tag{53}$$

We optimize the sixteen angles jointly across training conditions and keep them fixed during conditional generation. These local rotations act on a correlated input and can therefore produce correlated output distributions without learned entangling gates. Because the angles can differ between qubits, the learned layer need not preserve the input's permutation symmetry. We therefore evaluate the full distribution to capture discrepancies among bitstrings with the same Hamming weight.

The reference state $H^{\otimes 16}|\phi(u,v)\rangle$ retains the input's permutation symmetry, so its amplitudes depend only on Hamming weight. This structure enables efficient classical sampling and amplitude-ratio evaluation for training. Appendix B describes these operations and the resources required to prepare the correlated input.

We compare two training designs, each with nine fitting conditions and 1024 target samples per condition. In the *separate-variation* design, we vary one coordinate at a time: u at $v = 0$, and v at $u = 0$. We exclude all pairs with both coordinates nonzero from fitting and development, allowing us to test generalization to unseen joint combinations. In the *joint-variation* design, we fit the nine points of the 3×3 grid $\{-1, 0, 1\}^2$.

For each design, we select hyperparameters using exact full-distribution TVD at development conditions and perform five independent fits with the selected configuration. Within each fit, we retain the parameters that minimize development TVD. Appendix B specifies the condition sets, optimization settings and selection procedure.

The separate-variation model generalizes to sixteen joint condition pairs excluded from fitting and development, achieving mean full-distribution TVD 0.0637 ± 0.0051. Four pairs lie inside the convex hull of its fitting conditions and give mean TVD 0.0311 ± 0.0022; the remaining twelve lie outside and give 0.0745 ± 0.0061. All sixteen pairs combine coordinate values encountered separately during fitting, so this evaluation tests new combinations within the fitted marginal ranges. The joint-variation model achieves mean full-distribution TVD 0.0646 ± 0.0002 over thirty-two withheld reporting pairs. For each fit, we average TVD over the reporting pairs, then report the mean and sample standard deviation of these averages across five fits. Because the designs use different reporting sets, we compare them on a common evaluation grid below.

After model selection, we evaluate both models on a common 41×41 grid (Figure 3). For the averages in Table II, we exclude every pair used for fitting or development by either design, leaving 1652 common evaluation pairs. The separate-variation and joint-variation models achieve mean full-distribution TVDs of 0.0498 ± 0.0040 and 0.0533 ± 0.0003, respectively, compared with 0.3124 for the encoder alone. Both learned models therefore improve substantially on the distributions supplied by the fixed encoder across this domain. We interpret this post-selection evaluation as a descriptive comparison of the frozen models because earlier exploratory studies examined some reporting conditions.

We assess sensitivity to each conditioning variable by fixing one encoder coordinate at zero without retraining and retaining the original target conditions. For the separate-variation model, fixing $v = 0$ increases mean full-distribution TVD from 0.0637 to 0.0773, while fixing $u = 0$ raises it to 0.3851. These increases show that the trained model uses both inputs, although they do not establish accurate reproduction of the target width response.

We also repeat model selection and training with one angle shared across all sixteen rotations. This simpler model achieves similar TVD but responds much less to the width coordinate. We therefore retain sixteen independent angles to preserve the stronger response to both conditioning variables. Appendix B details the simplification study and conditional-response diagnostics. Together, these results demonstrate two-dimensional conditional generation with a shared local rotation layer acting on a structured correlated input.

C. Curie-Weiss spin model

We use the zero-field Curie–Weiss model [42] to test whether the architecture captures changes in distributional shape. The spins $s_i \in \{-1, +1\}$ interact with equal strength across all pairs, so the target assigns equal probability to bitstrings with the same Hamming weight. As the inverse temperature β increases, the finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode the inverse temperature using the fixed product-state encoder

$$|\phi(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.75\beta) R_Y\left(\frac{\pi}{2}\right)|0\rangle_i . \tag{54}$$

The R_Z rotations encode β in the input phases while leaving the computational-basis measurement distribution uniform. We parameterize the IQP unitary with all one- and two-body generators, giving 136 trainable parameters. We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and evaluate at the held-out inverse temperatures $\beta \in \{0.4, 1.0, 1.6\}$.

We compare the shared conditional IQP model with two independently tuned baseline families. Each family comprises three circuits, which we fit separately at the three evaluation temperatures. The independent conditional IQPs retain the corresponding temperature encoder, whereas the independent unconditional IQPs use $|0\rangle^{\otimes 16}$ as their input state.

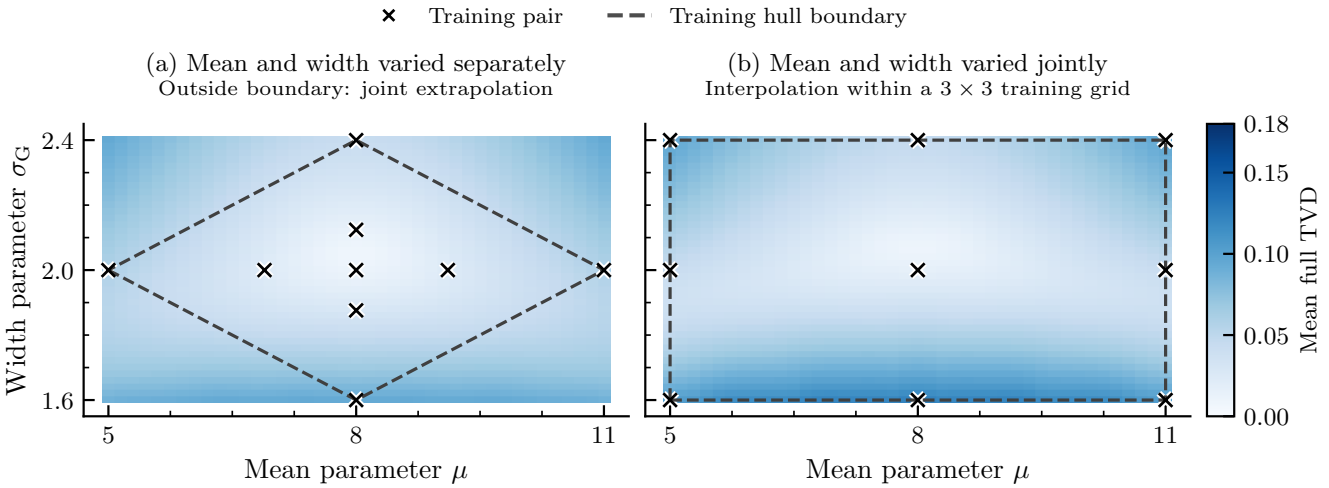


FIG. 3. Generalization across Gaussian profile centers and widths with a correlated encoder and sixteen learned local rotations. Colors show exact full-distribution total variation distance (TVD), averaged over five independent fits at each point of a 41×41 grid. Crosses mark fitting conditions, and dashed lines bound their convex hulls. (a) We vary center and width separately during fitting. Pairs outside the diamond test extrapolation beyond the convex hull of the separate-variation fitting conditions while remaining within each coordinate's fitting range. (b) We fit a 3×3 grid, whose interior tests interpolation. The axes show the target profile parameters μ and σ_G, and both panels use the same color scale. We evaluate the grid after completing model selection. Table II reports errors on a common domain excluding both designs' fitting and development pairs and includes the encoder-only baseline.

TABLE II. Full-distribution total variation distance (TVD) on a common set of Gaussian condition pairs. We exclude both designs' fitting and development pairs, leaving 1652 evaluation pairs. We define the inside and outside groups using the separate-variation fitting hull, $|u| + |v| \leq 1$, and apply this grouping to every model. For each trained model, we average TVD over conditions within each fit, then report the mean and sample standard deviation across five fits. The encoder-only baseline is deterministic. We evaluate the frozen models after selection.

	All unseen pairs 1652 pairs	Inside hull 828 pairs	Outside hull 824 pairs
Encoder only	0.3124	0.2215	0.4038
Mean and width varied separately	0.0498 ± 0.0040	0.0357 ± 0.0029	0.0640 ± 0.0052
Mean and width varied jointly	0.0533 ± 0.0003	0.0379 ± 0.0004	0.0688 ± 0.0002

We use exact full-distribution TVD for hyperparameter optimization (HPO) and model selection. We tune the shared model on development temperatures and each independent family jointly across its fitted temperatures. Appendix A 3 a details the protocol.

Each independent baseline family uses three circuits and therefore three times the total parameters and optimizer updates of the shared model. The shared model uses more training-pool samples, with 4,096 entries per run compared with 3,072 for each independent family. Appendix A 3 reports the detailed budgets.

The shared conditional model captures the change from unimodal to bimodal magnetization profiles, including bimodality at the held-out inverse temperature $\beta = 1.0$ (Figure 4). It achieves lower mean full-distribution TVD than both independent baseline families at $\beta = 1.0$ and 1.6, with mean errors of 0.2175 and 0.1709, respectively (Table III). The temperature $\beta = 1.6$ lies beyond the training range. At $\beta = 0.4$, the independent conditional family achieves lower mean TVD than the shared model (0.1240 versus 0.2240), while the uniform encoder-only baseline achieves the lowest error (0.1180). Together with the held-out results, this comparison suggests that learning across conditions may provide a useful inductive bias for generalization.

The profile diagnostic in Appendix A 3 (Table VIII) gives a different ranking from full-distribution TVD. At $\beta = 1.0$, both independent baseline families achieve lower mean profile TVD than the shared model but higher mean full-distribution TVD. At $\beta = 0.4$, the shared model reduces mean profile TVD relative to the encoder while increasing mean full-distribution TVD. These differences show that agreement in Hamming-weight profiles does not ensure accurate bitstring probabilities.

We also evaluate physical observables to distinguish magnetization magnitude from energy fluctuations. The exact zero-field target has $\langle m \rangle = 0$ at finite size, so we report the mean absolute magnetization $\langle |m| \rangle$, with

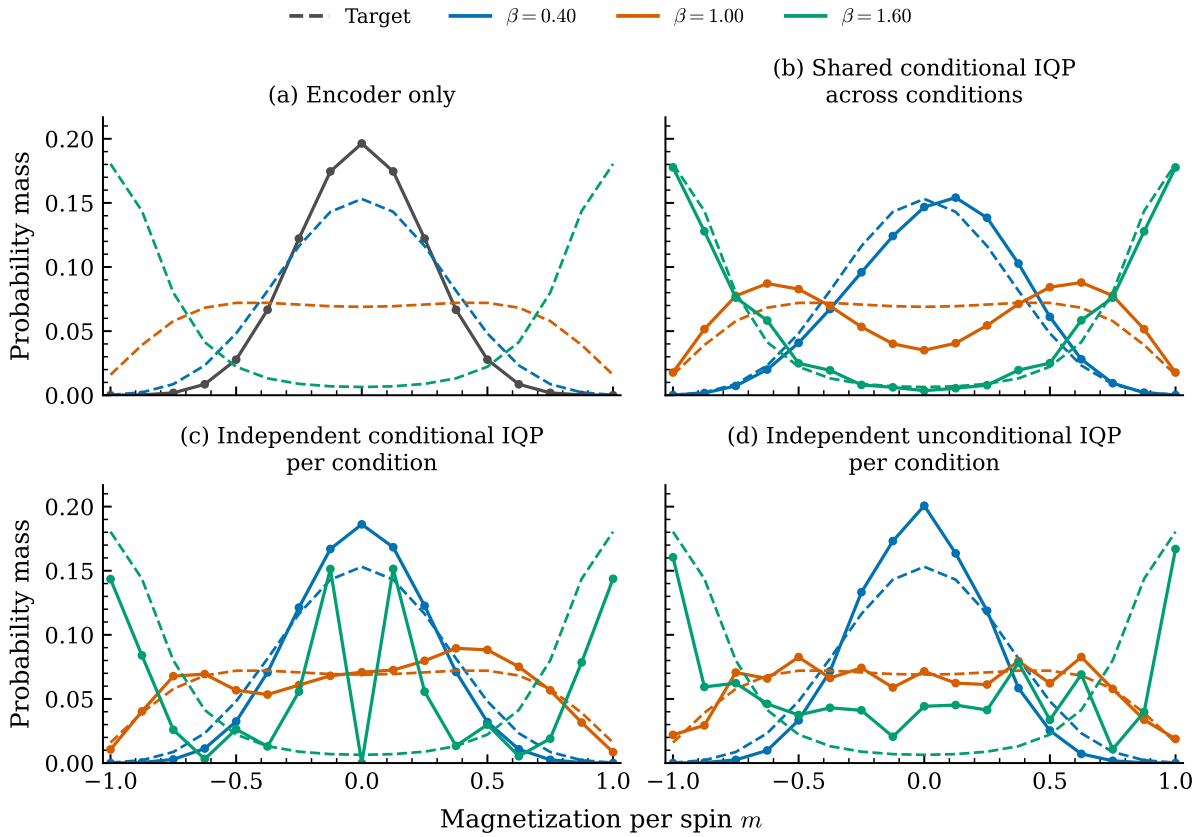


FIG. 4. Curie–Weiss magnetization profiles at sixteen qubits. We write the magnetization as $m = (2k - 16)/16$, where k is the Hamming weight. Panels show (a) encoder only, (b) shared conditional IQP, (c) independent conditional IQPs, and (d) independent unconditional IQPs. Colors distinguish $\beta = 0.4, 1.0, 1.6$, and dashed curves show exact targets. Solid curves show exact model profiles averaged over five confirmation runs used for model selection and reporting; the encoder-only profile is deterministic. We evaluate the shared model at temperatures excluded from fitting and model selection, whereas we fit the independent circuits directly at each displayed temperature. Table III reports full-distribution TVD.

TABLE III. Exact full-distribution TVD for sixteen-qubit Curie–Weiss models. For trained models, we report means ± sample standard deviations across five confirmation runs used for model selection and reporting. The encoder-only baseline is deterministic. We evaluate the shared model at temperatures excluded from fitting and model selection, whereas we fit the independent circuits directly at each evaluation temperature. Bold entries identify the lowest mean TVD in each column, including the encoder-only baseline.

Model	Evaluation	Full-distribution TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	**0.1180**	0.4406	0.8699
Shared conditional IQP	Held-out	0.2240 ± 0.0115	$\mathbf{0.2175 \pm 0.0136}$	$\mathbf{0.1709 \pm 0.0136}$
Independent conditional IQPs	Fitted	0.1240 ± 0.0083	0.2309 ± 0.0319	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.3842 ± 0.0017	0.3322 ± 0.0105	0.3299 ± 0.0119

$m = n^{-1}\sum_i s_i$ and $n = 16$. For the Hamiltonian $H = -nm^2/2$, we define the per-spin heat-capacity fluctuation estimate

$$c_{\mathrm{fluc}}(\beta) = \frac{\beta^2}{n}\big(\langle H^2\rangle - \langle H\rangle^2\big). \tag{55}$$

This expression equals the per-spin equilibrium heat capacity for the Gibbs target. For a learned distribution, it measures energy fluctuations and need not equal the temperature derivative of the mean energy per spin. We compute both observables from exact probabilities for the same five selected confirmation circuits on a grid from $\beta = 0$ to 2 in steps of 0.02, without additional fitting or model selection. The sweep includes fitting and development temperatures as well as temperatures beyond

the fitting range; it is not an entirely held-out evaluation. Appendix A 3 b specifies the calculation and aggregation.

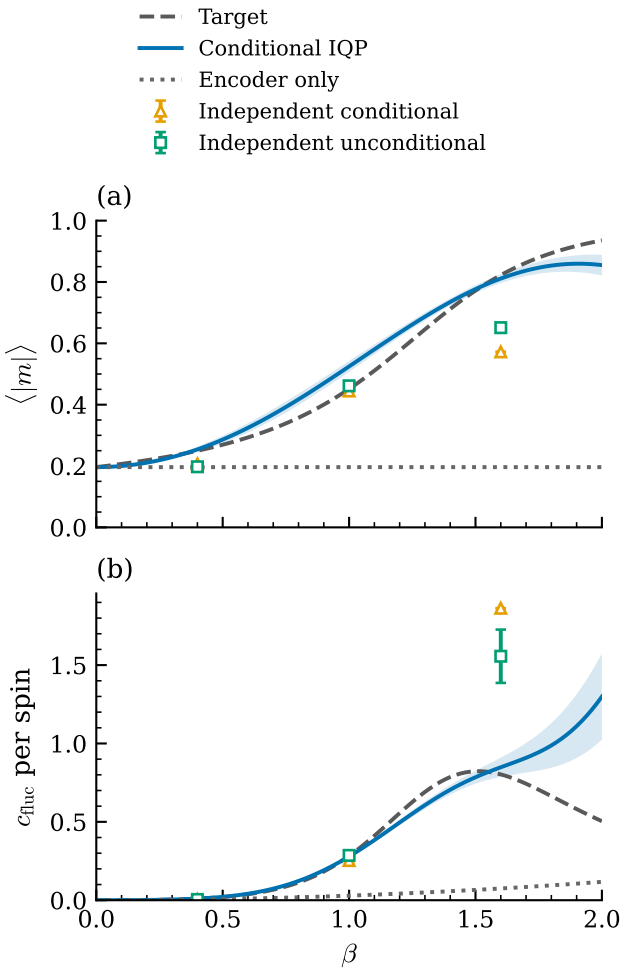


FIG. 5. Sixteen-qubit Curie–Weiss observables: (a) mean absolute magnetization and (b) per-spin heat-capacity fluctuation estimate, Eq. (55). Dashed and dotted curves show the exact target and uniform encoder reference. Conditional IQP curves and bands show mean and sample standard deviation across five frozen confirmation circuits at 101 temperatures. Independent markers use separately fitted circuits at $\beta = 0.4, 1.0, 1.6$, with the same summaries. We compute nonlinear observables within each run before aggregation. The sweep includes fitting, development and extrapolation conditions. For learned distributions, panel (b) measures energy fluctuations rather than a verified temperature response.

The dense response reveals errors that the three-temperature distribution comparison does not fully describe (Figure 5). At $\beta = 1.0$, the shared model overestimates mean absolute magnetization (0.525 versus the target 0.451), while the independent fitted baselines are closer. At higher inverse temperatures, the shared response levels off and underestimates the target (0.855 versus 0.936 at $\beta = 2$). Energy fluctuations show a different discrepancy. The target's heat capacity reaches its maximum on the evaluated grid near $\beta = 1.52$ and then decreases, whereas the shared model's fluctuation estimate continues increasing through $\beta = 2$, reaching 1.299 versus the target 0.504. At $\beta = 1.6$, however, the shared estimate remains closer to the target than either independent baseline. Thus, capturing the change in magnetization magnitude does not ensure recovery of the target's energy fluctuations, and the relative accuracy depends on the observable and temperature.

D. Conditional generation with exact switching symmetry

We use a signed-graph family to demonstrate how a correlated input can impose exact symmetry throughout conditional learning. Sixteen qubits represent the edges of a seven-vertex graph. The graph contains the complete graph on vertices $0, \ldots, 4$ and connects each of vertices 5 and 6 to vertices $0, 1, 2$. For edge bits x_{ij} and signs $s_{ij} = (-1)^{x_{ij}}$, we define

$$p_\beta(x) = Z_\beta^{-1} e^{\beta E(x)}, \qquad E(x) = \sum_{\{i,j,k\}\in\mathcal{T}} s_{ij} s_{ik} s_{jk}, \tag{56}$$

where $\mathcal{T}$ contains the graph's sixteen triangles. Increasing β favors positive triangle products. A vertex switch replaces each edge sign by $h_i s_{ij} h_j$, with $h_i \in \{-1, 1\}$. Each triangle contains two edges incident on each of its vertices, so every triangle product remains unchanged. Fixing $h_0 = 1$ gives a group G of sixty-four distinct switching masks.

We encode temperature through the graph's fundamental triangles. The six edges incident on vertex 0 form a spanning tree. Each of the ten remaining edges, called chords, completes a triangle $(0, i, j)$. Starting from $|0\rangle^{\otimes 16}$, we rotate every chord by $R_X(t_\beta)$ and apply two CNOTs from that chord to edges $0i$ and $0j$. Figure 6 shows this motif. The CNOTs combine the triangle bits so that every vertex has even incidence parity. After all motifs, we apply one global Hadamard layer. The resulting input is

$$|\phi_\beta\rangle = \exp\left[-\frac{i t_\beta}{2} \sum_{(i,j)\in\mathcal{C}} Z_{0i} Z_{0j} Z_{ij}\right] |+\rangle^{\otimes 16}, \tag{57}$$

$$t_\beta = \frac{\pi}{2} \tanh(1.5\beta), \tag{58}$$

where $\mathcal{C}$ is the chord set. The preparation uses ten rotations, twenty CNOTs and sixteen Hadamards, without ancillary qubits. Its computational-basis probabilities are uniform; the temperature dependence enters through correlated phases. Ten independent Bernoulli variables and their cycle embedding provide direct sampling and amplitude-ratio access for the pre-Hadamard reference. Appendix C gives these operations explicitly.

The input enforces switching symmetry independently of the learned parameters. Even incidence parity implies

$X^g |\phi_\beta\rangle = |\phi_\beta\rangle$ for every $g \in G$. Any IQP layer generated by X strings commutes with these switches. We use the particularly simple learned layer

$$U_{\alpha,\gamma} = \exp\left(i\alpha \sum_{i=1}^{6} X_{0i} + i\gamma \sum_{(i,j)\in\mathcal{C}} X_{ij} \right), \tag{59}$$

with one angle for the root edges and one for the chords, shared across all temperatures. This layer adds sixteen local rotations and no entangling gates. Its Born distribution obeys

$$q_{\alpha,\gamma}(x \mid \beta) = q_{\alpha,\gamma}(x \oplus g \mid \beta) \tag{60}$$

throughout optimization and at unseen conditions, without averaging the output over switches.

Finite-data training produces a conditional response with moderate full-distribution error. We fit at $\beta \in \{0, 0.4, 0.8, 1.2\}$ using 128 target bitstrings per temperature, or 512 target samples per fit. Sampled MMD drives optimization; exact development distributions determine hyperparameters and checkpoints. After freezing the configuration, we train five independent final fits. Their mean full-distribution TVD across twelve reporting temperatures is 0.1780 ± 0.0013, with mean per-fit worst-temperature TVD 0.2981 ± 0.0018. Table IV separates interpolation from extrapolation. These errors refer to the full sixteen-qubit distribution, which we evaluate exactly through its switching quotient. The reported spreads are sample standard deviations across the five fits.

We describe conditional graph order using mean triangle balance $b = \langle E \rangle / 16$ and fully balanced probability $P_{\text{bal}} = \Pr(E = 16)$. The latter measures the probability that all sixteen triangle products are positive. Figure 7 shows both observables for the frozen fits over $\beta \in [0, 2.4]$, with the encoder-only uniform distribution as a reference. The learned response increases with inverse temperature and then saturates below the target. At $\beta = 2.4$, the model gives $b = 0.9148 \pm 0.0017$ and $P_{\text{bal}} = 0.7983 \pm 0.0046$, compared with target values 0.9999 and 0.9996. The dense sweep is a post-selection diagnostic that includes fitting and development temperatures. A separate extension to ten temperatures from 1.5 to 2.4 gives mean TVD 0.1981 ± 0.0046.

The experiment demonstrates tractable correlated conditioning and exact architectural symmetry with two learned angles. The encoder alone has mean reporting TVD 0.7275; holding the selected models' input temperature fixed at 0.6 raises mean TVD to 0.4886 ± 0.0011. This sensitivity check supports a role for the conditional input, but does not compare against an optimized temperature-blind model. The selected order observables also do not establish complete thermodynamic fidelity: energy fluctuations remain inaccurate at large β, and a fixed classical switching control achieves lower reporting TVD. Appendix C reports both diagnostics. The construction therefore supports the framework's symmetry capability without establishing a classical performance advantage.

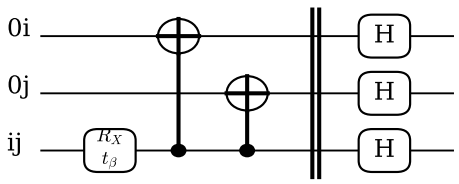


FIG. 6. Preparation motif for one fundamental triangle. A chord rotation $R_X(t_\beta)$ and two CNOTs prepare $\cos(t_\beta/2)|000\rangle - i\sin(t_\beta/2)|111\rangle$ on edges $0i, 0j, ij$. The full encoder repeats the rotation and CNOT construction for ten chords with shared root edges, then applies the global Hadamard layer once. The displayed motif is therefore an illustration, rather than an independent three-qubit block repeated with its own Hadamards. Every rotation uses $t_\beta = (\pi/2)\tanh(1.5\beta)$.

TABLE IV. Exact full-distribution TVD of five sixteen-qubit fits with 128 target samples per fitting temperature. Entries show mean ± sample standard deviation across per-fit temperature averages. The worst-condition row averages each fit's maximum TVD. The original reporting conditions do not enter selection for this graph. The last row describes a separate post-selection range extension.

Reporting group	TVD ↓
Original report, 12	0.1780 ± 0.0013
Interpolation, 10	0.1775 ± 0.0006
Extrapolation, 2	0.1806 ± 0.0046
Worst condition	0.2981 ± 0.0018
Extension 1.5–2.4, 10	0.1981 ± 0.0046

E. Scaling

We assess the computational feasibility of classical training at $n \in \{64, 128, 256, 512, 1024, 2048\}$ visible qubits, without ancillas. We use the fixed product-state encoder $R_Z(0.75\beta)R_Y(\pi/2)$ on every qubit. The learned unitary uses independent one-body phase gates and two-body phase gates between horizontal and vertical nearest neighbors on an open rectangular lattice. For an $L_x \times L_y$ lattice with $n = L_x L_y$, the model has $3n - L_x - L_y$ trainable parameters, so the parameter count grows linearly with n. Appendix A 4 gives the lattice dimensions and optimization settings.

At each system size, we train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and track the mean sampled MMD^2 over the held-out temperatures throughout optimization (Figure 8). Appendix A 4 specifies the training and reporting protocol.

The scaling study demonstrates the computational feasibility of classical conditional training up to 2048 qubits

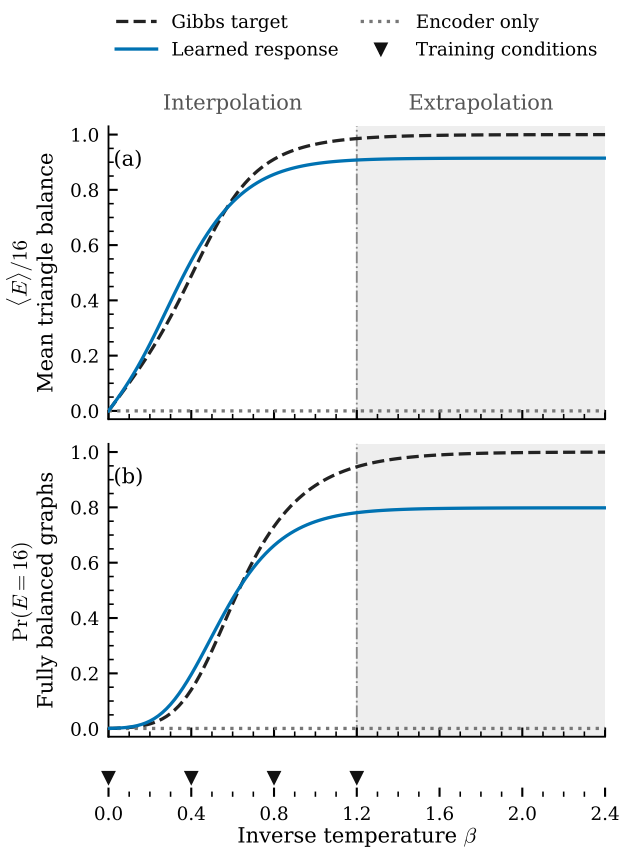


FIG. 7. Conditional graph order with exact switching symmetry on sixteen qubits. (a) Mean triangle balance $\langle E\rangle/16$. (b) Probability $\Pr(E=16)$ that all sixteen triangles are balanced. Solid blue curves show the mean across five frozen fits, and bands show one sample standard deviation. Each fit uses 128 target bitstrings at each of four fitting temperatures. Dashed black curves show the exact Gibbs target; dotted gray curves show the encoder-only uniform distribution. Downward triangles mark fitting temperatures, and shading marks extrapolation beyond $\beta = 1.2$. We evaluate 121 points over $\beta \in [0, 2.4]$ after model selection, including fitting and development conditions. The learned response increases with inverse temperature but saturates below the target.

(Figure 8). At 64 qubits, the median held-out sampled MMD^2 decreases from 0.0504 to 0.0350 after 5,000 updates; at 2048 qubits, it decreases from 0.0490 to 0.0417. These modest reductions suggest limited progress under the chosen training configuration and may reflect suboptimal hyperparameters. The runs use a sparse lattice ansatz and optimizer settings that differ from the tuned sixteen-qubit comparison. Further tuning and benchmarking are needed for a systematic assessment of generative performance. Because the kernel bandwidths vary with system size, the absolute MMD^2 values do not directly compare distributional fidelity across sizes.

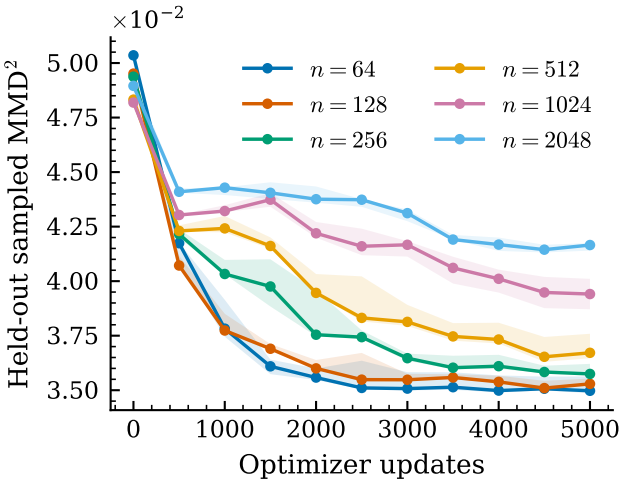


FIG. 8. Classical training of conditional IQP Born machines from 64 to 2048 qubits. At each plotted update, we average sampled MMD^2 over the held-out inverse temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. Curves and shaded bands show the median and interquartile range of these averages across three independent training runs. We fix the expected mask weights at $w \in \{2, 4\}$ for each system size, so the corresponding kernel bandwidths vary with n.

V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

Input-state encoding can also enforce exact output symmetries without restricting the learned phases. For a nonzero bit mask $s \in \{0,1\}^n$, the bit-flip operator $X_s = \bigotimes_{i=1}^n X_i^{s_i}$ commutes with the IQP unitary. If each encoded input is an eigenstate of X_s, the output state retains the same eigenvalue. Its computational-basis amplitudes at x and $x \oplus s$ therefore differ at most by a sign, giving equal probabilities for every condition and parameter vector. This applies the commuting-symmetry construction of Recio-Armengol *et al.* [28, Sec. 6] to conditional inputs. Such constraints provide inductive biases in variational quantum models [43].

The conditional switching experiment in Section IV D implements this criterion for six independent masks simultaneously. Its reference state has even incidence parity at every graph vertex. After the Hadamards, the input is a simultaneous $+1$ eigenstate of all vertex switches, so any subsequent X-generated IQP layer preserves the constraints. Parameter tying reduces the fitted model to two angles but is not required for the symmetry guarantee. This separates an architectural property from distributional accuracy, which still depends on the target, encoder and optimization.

VI. DISCUSSION

Our framework establishes sufficient conditions for classical training of conditional IQP Born machines with input preparation beyond strict IQP form. Efficient sampling and amplitude-ratio evaluation of the reference state enable unbiased estimation of conditional training moments and their gradients, while one learned unitary shares parameters across conditions. The sixteen-qubit experiments demonstrate conditional translations and changes in distributional shape through input phases and correlated amplitudes. The two-dimensional Gaussian experiment also approximates mean–width combinations absent from fitting and development, including pairs outside the joint training hull of separately varied coordinates. For Curie–Weiss, the shared model achieves lower mean full-distribution TVD than both independent baseline families at $\beta = 1.0$ and 1.6, with the latter extending beyond the training range. Each baseline family fits these temperatures directly and uses three times the aggregate parameters and optimizer updates, whereas the shared model uses more training-pool samples. At $\beta = 0.4$, the uniform encoder-only distribution achieves the lowest error among the models in the primary comparison. Joint training may provide useful regularization, but comparisons with matched data and optimization budgets are needed to assess this explanation.

Input-state design can impose exact output symmetries while preserving classical trainability. The conditional signed-graph construction guarantees sixty-four switching translations for every learned parameter vector. Five independent finite-data fits give mean reporting TVD 0.1780 ± 0.0013 with two learned angles, while the graph-order response extends beyond the fitting range. This symmetry guarantee does not ensure a faithful Gibbs response: the learned balance saturates below the target and high-inverse-temperature energy fluctuations remain inaccurate. A fixed classical switching control achieves lower reporting TVD, and the model admits exact quotient evaluation. These results support correlated conditional encoding and symmetry control without establishing a classical fidelity or computational advantage.

The methodological limitations concern distributional evaluation and the scope of the scaling evidence. Efficient, unbiased MMD estimation does not ensure accurate recovery of every relevant correlation [44]. Although the Gaussian kernel includes all correlation orders, its bandwidth weights them unequally, and finite sampling and optimization may fail to resolve weakly weighted discrepancies. Agreement in Hamming-weight profiles can therefore coexist with errors in individual bitstring probabilities, making full-distribution diagnostics necessary whenever feasible. The observable sweep further shows that recovering magnetization trends does not ensure accurate energy fluctuations, especially beyond the fitting range. For Curie–Weiss, reporting reuses the confirmation runs used to select hyperparameters, so the reported spread does not independently assess model selection. The reporting temperatures are excluded from fitting and selection but were examined in earlier experiments. Independent final fits and previously unexamined conditions, evaluated with configurations fixed in advance, would strengthen the generalization evidence. The sparse-lattice runs establish computational feasibility up to 2048 qubits, with modest reductions in held-out sampled MMD^2. Their connectivity and optimizer settings differ from the sixteen-qubit comparison. Fixed expected mask weights emphasize low-order correlations as the system grows, while size-dependent bandwidths prevent direct comparisons of distributional accuracy across sizes.

Future work should examine richer tractable encoders and optimization strategies that improve conditional accuracy. Broader families of correlated inputs and multidimensional encodings should be assessed alongside state-preparation and estimation costs, with particular attention to calibration of each conditional response. Conditional extensions of spectral Born machines would require corresponding sampling, amplitude-ratio and gradient estimators [45]. Generator selection, initialization and adaptive kernels merit investigation [39–41], including whether gradients from different conditions reinforce or cancel one another. Quantum deployment should assess how noise and circuit partitioning affect the conditional response and imposed symmetries [46, 47]. Practical quantum sampling advantage remains open: our Gaussian, Curie–Weiss and signed-graph targets admit efficient classical sampling. Demonstrations would require comparisons with strong classical generative models and sampling-hardness or verification arguments appropriate to the trained conditional circuits.

VII. CONCLUSIONS

We introduced a conditional framework for IQP Born machines that combines a shared learned unitary with encoders that need not have IQP form. Efficient reference-state sampling and amplitude-ratio evaluation enable unbiased classical estimation of conditional training observables and their gradients, allowing optimization without quantum circuit sampling. Input-state design also lets us impose exact output symmetries independently of the learned parameters.

Our sixteen-qubit experiments demonstrate conditional translations and changes in distributional shape through input phases and correlated amplitudes. The two-dimensional Gaussian model generalizes to joint mean–width combinations absent from fitting and development using sixteen learned local rotations. For the Curie–Weiss model, the shared model achieves lower mean full-distribution TVD than both independently fitted baseline families at two held-out temperatures, including one beyond the training range, with fewer aggregate parameters and optimizer updates. The signed-

graph experiment demonstrates finite-data conditional generation with exact switching symmetry and two learned angles. Additional runs establish the computational feasibility of classical conditional training up to 2048 qubits. Together, the theoretical constructions and numerical results establish conditional generation and symmetry control within a framework that supports classical training. Further work must assess distributional accuracy at larger scales and practical quantum advantage.

DATA AVAILABILITY

The code and data supporting this study are available on GitHub [48].

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Appendix A: Hyperparameters and evaluation protocols

1. Common training settings

We train all models using the unbiased estimator of the squared-MMD objective. We keep the encoders fixed and initialize the circuit phases independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. At each update, we sample one fitting condition and one kernel bandwidth uniformly, then draw a target minibatch with replacement. We draw fresh nonidentity masks and apply the probability correction for excluding the identity mask. Mask budgets count draws, including repeated masks. We draw reference samples independently of the target minibatch and reuse them across masks. Table V summarizes the translation, Curie–Weiss and scaling settings. Appendices B and C give the separate two-dimensional Gaussian and conditional switching protocols.

For each initialization, we reset the optimizer and draw new circuit phases while reusing the target pools. For the experiments summarized in Table V, each training run includes all prescribed initializations, and we select the checkpoint with the lowest selection score across them. Reported uncertainty summarizes variation across independent training runs.

We train the Gaussian and scaling models with `Adam` [49], using cosine learning-rate decay to one tenth of the initial rate. For the Curie–Weiss models, we use the optimizer selected by HPO and keep the learning rate constant. We set $b_1 = 0.9$, and use $b_2 = 0.999$ and $\epsilon = 10^{-8}$ for the `Adam` variants. We set weight decay to zero.

TABLE V. Training settings and budgets per circuit. Mask and reference budgets give the number of draws per update. We select Curie–Weiss hyperparameters and checkpoints using full-distribution TVD. Table VII gives the search space for the optimizer, learning rate and initialization width. We report selected learning rates and initialization widths to three significant figures and display the expected mask weight 7.99 as 8. For Curie–Weiss, the update and initialization budgets apply to confirmation and evaluation; screening uses the smaller budgets specified in Sec. A 3 a.

Setting	Gaussian translation	Shared product input	Independent conditional	Independent unconditional	Scaling
Updates per initialization	1,024	2,000	2,000	2,000	5,000
Initializations per run	1	3	3	3	3
Reported runs	5	5	5	5	3
Training targets per condition	512	1,024	1,024	1,024	10,000
Target minibatch	128	256	256	256	450
Training mask draws	1,024	2,048	2,048	2,048	512
Training reference draws	1,024	2,048	2,048	2,048	1,024
Expected mask weights	{1}	{2, 4, 8}	{2, 4, 8}	{2, 4, 8}	{2, 4}
Checkpoint interval	32	100	100	100	25
Optimizer	`Adam`	`Adam`	`AdamW ScheduleFree`	`Adam`	`Adam`
Learning rate	0.005	0.0310	0.0603	0.00844	0.01
Learning-rate schedule	Cosine	Constant	Constant	Constant	Cosine
Initialization width s_θ	0.25	0.696	0.119	0.117	0.1
Coefficient b_1	0.9	0.9	0.9	0.9	0.9

For an expected mask weight w, we set the Gaussian bandwidth to

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n denotes the number of visible qubits and w is the expected mask weight before excluding the identity mask. We weight the bandwidths equally in the objective. For the sixteen-qubit Curie–Weiss models, we use $w \in \{2, 4, 7.99\}$ and display 7.99 as 8. Using 7.99 keeps $w < n/2$ and the bandwidth strictly positive.

We report rotation angles in radians and compute sixteen-qubit full and profile TVDs from exact circuit probabilities. Sampled-MMD diagnostics do not enter these calculations.

2. Discrete Gaussian translation

We use 16 visible qubits on an open 4×4 lattice. The learned unitary contains 16 one-body phase gates and 24 two-body phase gates between horizontal and vertical nearest neighbors. We prepare the input state with the fixed product-state encoder

$$|\phi_{\mathrm{G}}(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c-1/2)R_Y(1.5)\,|0\rangle_i\,. \tag{A2}$$

We choose the near-equatorial angle 1.5 radians empirically to avoid an input state whose output distribution the learned unitary cannot change. At $c = 0.5$, replacing this angle with $\pi/2$ would prepare $|+\rangle^{\otimes 16}$, a simultaneous eigenstate of every X-type generator in the IQP unitary. The learned parameters would then contribute only a global phase, leaving a uniform output distribution with a fixed binomial Hamming-weight profile. Using 1.5 removes this exact invariance and allows the learned parameters to modify the output distribution at $c = 0.5$.

We use the target Hamming-weight profile $\rho(k \mid c) \propto \exp[-(k-(5+6c))^2/8]$, where $k = |x|$. Within each Hamming-weight shell, we assign equal probability to all bitstrings. We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$, using 512 target samples per condition and 2560 per run.

We evaluate the model at initialization and every 32 updates through update 1024, selecting the checkpoint with the lowest mean exact profile TVD over the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. We then evaluate the selected model at 80 held-out conditions, $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$. We draw 1000 target samples per evaluation condition for auxiliary sampled-MMD diagnostics.

We compute full-distribution and profile TVDs from exact circuit probabilities. For each run, we average each error over the evaluation conditions and report the median of these averages across five independent runs. Figure 2 averages exact profiles across the five selected circuits at the development conditions and shows the median and interquartile range of their mean output Hamming weights at each held-out condition.

3. Curie–Weiss models

We use the sixteen-qubit target $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. For all trained sixteen-qubit Curie–Weiss models, we parameterize the learned unitary with 16 one-body phase gates and 120 two-body phase gates, without ancillas. For the shared product-input model, we prepare $R_Z(0.75\beta)R_Y(\pi/2)\,|0\rangle$ on every qubit.

We train the shared product-input model at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, use $\beta \in \{0.5, 0.7, 1.3, 1.5\}$ for model selection, and report performance at $\beta \in \{0.4, 1.0, 1.6\}$. Each independent baseline family comprises three circuits that we fit directly at these reporting temperatures, using either the corresponding product input or $|0\rangle^{\otimes 16}$. For each baseline family, we select one optimizer, learning rate and initialization width jointly for its three circuits.

We train each circuit using 1024 target samples per fitting temperature and run, minibatches of 256, and 2048 mask draws and 2048 reference draws per update. Table VI summarizes the aggregate fitting resources.

For auxiliary sampled-MMD diagnostics, we use 2048 validation targets per fitting temperature and 2048 independently drawn test targets per reporting temperature. Each diagnostic uses 2048 mask draws per condition and bandwidth, and 2048 reference samples per condition shared across bandwidths. We do not use these diagnostics for model selection.

TABLE VI. Curie–Weiss fitting budgets per run for confirmation and evaluation. Training-target counts sum the pool entries over fitting temperatures and exclude repeated minibatch draws. Optimizer-update counts include all three initializations of every circuit. The shared model fits one circuit, and each independent baseline family fits three.

Model	Trained circuits	Training targets	Optimizer updates
Shared product input	1	4,096	6,000
Independent conditional	3	3,072	18,000
Independent unconditional	3	3,072	18,000

a. *Hyperparameter optimization*

We report one Optuna study for each of the three Curie–Weiss model families, using a tree-structured Parzen estimator (TPE) [50]. Each study evaluates 60 configurations. We tune the optimizer choice among `Adam`,

`AdamWScheduleFree` and `SGDScheduleFree`, the constant learning rate, and the Gaussian initialization width [49, 51]. Table VII gives the search space. We assign each family the same number of trials, although each independent-family trial fits three circuits per run.

We use random search as a reference for the trial budget [52]. For 60 independent configurations drawn from a fixed sampling distribution, the probability of sampling at least one configuration in its best-performing 5% is $1 - 0.95^{60} \approx 0.954$. This calculation motivates the budget but does not establish equivalent coverage for adaptive TPE proposals.

TABLE VII. Curie–Weiss hyperparameter search space. We treat optimizer choice as categorical and search learning rates and Gaussian initialization widths on a logarithmic scale.

Hyperparameter	Options or range
Optimizer	`Adam`, `AdamWScheduleFree`, `SGDScheduleFree`
Learning rate, `Adam` variants	$[3 \times 10^{-4}, 10^{-1}]$
Learning rate, `SGDScheduleFree`	$[10^{-3}, 1]$
Initialization width s_θ	$[0.01, 0.7]$ radians

We begin each study with 12 startup trials, including one reference configuration per optimizer. We set the reference learning rates to 0.01 for the `Adam` variants and 0.1 for `SGDScheduleFree`, with $s_\theta = 0.5$ for all reference configurations. We do not prune trials.

We fix the optimizer coefficients as specified in the common training settings and use no warmup or gradient clipping. For schedule-free optimizers, we set the learning-rate weighting exponent to 2 and score checkpoints using the averaged parameters provided for evaluation.

For each screening trial, we train every circuit for 1000 updates on each of three runs, using one initialization per run. We compute exact TVD from the full distribution over the 2^{16} computational-basis states at initialization and every 100 updates. For each shared model, we select the checkpoint with the lowest mean TVD over the development temperatures. Within each independent baseline family, we select each circuit's checkpoint using TVD at its fitted temperature and average the three selected errors within each run. We then take the median of the three run scores as the trial score.

We confirm the three best distinct configurations from each study on five fresh runs, training each circuit for 2000 updates from each of three initializations per run. We select each circuit's checkpoint across its initializations using the same TVD criterion as in screening, then choose the configuration with the lowest median score across the five runs.

We report performance using the five selected confirmation runs, without further fitting or checkpoint selection. Because these runs also inform hyperparameter selection, they do not provide an independent final evaluation. In the current protocol, we exclude the shared models' reporting temperatures from fitting and selection, although earlier experiments examined them. We fit the independent circuits directly at their reporting temperatures. For each temperature, we report the mean and sample standard deviation across the five runs.

TABLE VIII. Exact Hamming-weight profile TVD for sixteen-qubit Curie–Weiss models. We report means ± sample standard deviations across the five confirmation runs used for model selection and reporting. We select models using full-distribution TVD; encoder-only values are deterministic. We evaluate the shared model at temperatures excluded from fitting and selection, and fit each independent circuit at its reporting temperature. Bold entries identify the lowest mean error at each temperature, including the encoder-only baseline.

Model	Evaluation	Hamming-weight profile TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	0.1180	0.4406	0.8699
Shared conditional IQP	Held-out	0.0976 ± 0.0269	0.1301 ± 0.0317	$\mathbf{0.0631 \pm 0.0135}$
Independent conditional IQPs	Fitted	$\mathbf{0.0939 \pm 0.0073}$	0.0885 ± 0.0116	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.1186 ± 0.0040	$\mathbf{0.0780 \pm 0.0320}$	0.3293 ± 0.0119

Table VIII reports exact Hamming-weight profile errors for the primary Curie–Weiss comparison.

b. Physical observables

For a distribution q, let $\rho_q(k \mid \beta)$ denote its exact Hamming-weight profile and set $m_k = (2k-n)/n$ with $n = 16$. We compute its moments as $\langle m^r \rangle_q = \sum_{k=0}^{n} \rho_q(k \mid \beta) m_k^r$, and its mean absolute magnetization as $\sum_k \rho_q(k \mid \beta)|m_k|$. The target law corresponds to $H = -nm^2/2$ in units $J = k_B = 1$. Excluding self-interactions would add only a constant to H and would not change the energy variance. Equation (55) therefore becomes

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2 n}{4} \left(\langle m^4 \rangle_q - \langle m^2 \rangle_q^2 \right). \tag{A3}$$

The observables depend only on the shell profile and do not test uniformity within shells. For the Gibbs target, differentiating the mean energy per spin with respect to $T = \beta^{-1}$ gives this fluctuation expression. We do not assume that the learned family satisfies the same derivative identity.

We evaluate the five selected shared product-input circuits on 101 equally spaced inverse temperatures $\beta_j = 0.02j$, $j = 0, \ldots, 100$, with parameters fixed after selection. The grid includes fitting and development temperatures, as well as extrapolation below and above the fitting range. These are the same confirmation runs used for hyperparameter selection, so the sweep is not an independent assessment of model selection. We compute each observable within each run before taking its mean and sample standard deviation, including the nonlinear heat-capacity estimate. The independent baselines appear only at their fitted temperatures, with the same calculation and aggregation. The uniform encoder reference has the binomial shell law $\rho_0(k) = 2^{-n}\binom{n}{k}$.

4. Scaling

We use the product-state Curie–Weiss encoder on every qubit, with no ancillas, and adopt the open lattice architectures listed in Table IX. At size n, the target distribution is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$.

We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, using 10,000 target samples per condition and 40,000 per run. We evaluate at $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ and use three independent runs at each size. We use the cosine-decay `Adam` settings in Table V, with 5000 updates per initialization and three initializations per run. These settings differ from the selected sixteen-qubit configurations. We have not tested the transfer of those configurations or retuned the scaling models.

TABLE IX. Open rectangular lattice architectures for the scaling experiments. We assign one trainable phase to each qubit and one to each horizontal or vertical nearest-neighbor pair.

Visible qubits	Lattice	Trainable parameters
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

We select the checkpoint with the lowest mean sampled validation MMD^2 across training temperatures and bandwidths. We evaluate initialization and every 25 updates across three initializations. We exclude held-out temperatures from fitting and selection.

For each run, we report the selected initialization at update 0 and every 500 updates through update 5000. At each reporting update, we average five independent MMD^2 estimates over the evaluation temperatures and bandwidths. Each estimate uses 2000 target draws per condition, 512 mask draws per condition and bandwidth, and 1024 reference draws per condition. We reuse reporting sample sets across updates and share reference samples across bandwidths. Figure 8 shows the median and interquartile range across the three runs. We retain negative unbiased MMD^2 estimates.

Appendix B: Two-dimensional Gaussian encoder and evaluation protocol

1. Fitting, development and reporting conditions

For joint-variation training, fitting conditions are $\{-1, 0, 1\}^2$ and development conditions are

$$\{-1, -0.5, 0, 0.5, 1\} \times \{-0.75, -0.25, 0.25, 0.75\}.$$

Reporting uses

$$u \in \{-1, -0.83, -0.37, 0.19, 0.61, 1\},$$
$$v \in \{-1, -0.79, -0.31, 0.23, 0.67, 1\}.$$

We omit the four fitted corners from the original reporting average, retaining thirty-two held-out pairs.

For separate-variation training, we fit $(u, 0)$ at $u \in \{-1, -0.37, 0, 0.37, 1\}$ and $(0, v)$ at $v \in \{-1, -0.31, 0.31, 1\}$. Development uses $(u, 0)$ at $u \in \{-0.73, -0.17, 0.17, 0.73\}$ and $(0, v)$ at $v \in \{-0.69, -0.13, 0.13, 0.69\}$. The sixteen reporting pairs are

$$\{-1, -0.37, 0.37, 1\} \times \{-1, -0.31, 0.31, 1\}.$$

Every mixed reporting pair is absent from both fitting and development. The condition $|u| + |v| \leq 1$ identifies the four pairs inside the joint fitting hull; twelve pairs lie outside it. No coordinate exceeds its marginal fitting range. Some of these target conditions appeared in earlier exploratory studies; the reporting claim concerns pairs withheld from the present models' fitting and selection.

2. Sampled training and model selection

Each run has a fixed pool of 1024 target bitstrings per fitting condition. Every update samples one condition and one kernel bandwidth uniformly, draws a target minibatch of 256 with replacement, and draws 1024 reference samples and 1024 nonidentity masks. We refresh masks and reference samples each update and retain the identity-mask probability correction. Conditional on the target pool, the sampled loss targets its empirical distribution. Adam uses $b_1 = 0.9$, $b_2 = 0.999$, $\epsilon = 10^{-8}$ and zero weight decay. The learning rate follows cosine decay to one tenth of its initial value.

We independently screen eight predefined settings for each training design using 512 updates and checkpoints every 64 updates. The settings vary the initial learning rate, Gaussian phase-initialization scale and expected mask weights. We confirm the two candidates with lowest development full TVD using two independent confirmation runs of 2000 updates each, selecting the candidate with lower mean development full TVD. Joint variation selects learning rate 0.01, initialization scale 0.25 and expected mask weights $\{1, 2, 4\}$. Separate variation selects learning rate 0.03, initialization scale 0.5 and expected mask weights $\{2, 4\}$. Confirmation and final fits evaluate checkpoints every 100 updates, including initialization and the last update. Checkpoints minimize mean exact development full TVD.

The eight settings are: learning rate 0.005, initialization scale 0.25, expected mask weight $\{1\}$; $(0.03, 0.5, \{2, 4\})$; $(0.003, 0.25, \{2, 4\})$; $(0.01, 0.25, \{2, 4\})$; $(0.03, 0.25, \{2, 4\})$; $(0.01, 0.1, \{2, 4\})$; $(0.01, 0.25, \{1, 2, 4\})$; and $(0.01, 0.25, \{2, 4, 6\})$. The mask law has the specified expected Hamming weight before conditioning on a nonidentity mask.

Within each selection stage, we match target pools and training random draws across candidates of the same design. We fix both selected configurations before evaluating either reporting grid and train five final fits independently of configuration selection. Heatmaps show pointwise mean full-distribution TVD across these fits. For table summaries, we first average over conditions within each fit, then compute the mean and sample standard deviation across fits. We do not evaluate the TVD of a mixture of fitted models.

3. Dense-grid evaluation and aggregation

After completing selection and the original reporting, we evaluate the frozen models on $u, v \in \{-1, -0.95, \ldots, 0.95, 1\}$. The heatmaps include all 1681 pairs and mark the fitting conditions. The common table domain removes any pair present in either design's fitting or development set, leaving 1652 pairs. Of these, 828 satisfy $|u| + |v| \leq 1$ and 824 lie outside that hull. This grouping is defined by the separate-variation design and applied identically to both models and the baseline. The denser grid is a descriptive post-selection evaluation; it does not supply an additional independent training replicate or a new confirmatory reporting set. Heatmap cells show directly evaluated points without smoothing.

4. Simplification and conditional-response diagnostics

We compare forty-phase lattice and sixteen-phase local layers with reciprocal-first-power and reciprocal-squared width profiles. We also test one angle shared across all sixteen local rotations, retaining the same correlated prepa-

ration. Each candidate uses the screening, confirmation and final-fitting protocol described above. The shared-angle model reduces the learned parameter count to one while retaining sixteen physical rotations.

Hyperparameter selection uses exact development full TVD. The architecture comparison is exploratory and adaptive, so its reporting coordinates do not constitute an independent confirmatory test. We retain sixteen independent local phases because the shared-angle model weakens width sensitivity despite comparable full TVD. Width-response and input-clamp diagnostics assess the resulting conditional response.

For the retained model, the endpoint width-response ratio is 0.405 after separate-variation training and 0.126 after joint-variation training. We compute the change in generated standard deviation between $v = -1$ and $v = 1$ at each fixed u in the original reporting grid, divide by the corresponding target change, and average over those u values and fits. This diagnostic uses the full original grid, including the fitted corners of the joint-variation grid that we exclude from its reporting TVD. A ratio of one would match the target change. The single-angle model gives ratios 0.073 and 0.083, respectively. Its original reporting TVD is 0.0665 ± 0.0001 for separate variation and 0.0633 ± 0.0002 for joint variation. For separate variation, freezing $v = 0$ increases the single-angle error to 0.0689, compared with 0.0773 for the retained model. We evaluate clamping on each original reporting set without retraining. These diagnostics measure the fitted model's sensitivity, rather than the optimum attainable by a condition-blind model. Local nonmonotonic width responses remain in one of five retained separate-variation fits and are more pronounced after joint variation.

5. Reference access and input-state preparation

We write the encoder's computational-basis amplitude at weight k as

$$a_{u,v}(k) = \sqrt{\frac{r_v(k)}{\binom{16}{k}}}\exp[-iu(16-2k)/4]. \tag{B1}$$

The Hadamard transform preserves permutation symmetry. For a reference bitstring of weight j, its amplitude is

$$\psi_{u,v}(j) = 2^{-8}\sum_{k=0}^{16} a_{u,v}(k)\mathcal{K}_k(j), \tag{B2}$$

$$\mathcal{K}_k(j) = \sum_{\ell=0}^{k}(-1)^\ell \binom{j}{\ell}\binom{16-j}{k-\ell}, \tag{B3}$$

with binomial coefficients outside their valid ranges equal to zero. We normalize and tabulate these seventeen amplitudes. Sampling first draws j with probability $\binom{16}{j}|\psi_{u,v}(j)|^2$, then draws a uniform weight-j bitstring. For a mask b and reference sample z, the estimator obtains its amplitude ratio from the entries at $|z|$ and $|z \oplus b|$. It only divides by amplitudes on the sampled support. For general n, the same construction uses a polynomial-size shell table and transform.

The correlated preparation precedes the local R_Z encoder and learned R_X rotations. It contributes to the physical resources even though it introduces no learned parameters. One preparation route maintains the Hamming weight of the prepared prefix in a clean counter. Define the per-string probability $g(k) = r_v(k)/\binom{n}{k}$, the suffix sum $F(n,k) = g(k)$, and the recursion

$$F(i,k) = F(i+1,k) + F(i+1,k+1). \tag{B4}$$

At prefix length i and prefix weight k, a counter-controlled R_Y rotation with angle

$$2\arcsin\sqrt{\frac{F(i+1,k+1)}{F(i,k)}} \tag{B5}$$

prepares the next conditional bit amplitude. We update the counter after each visible qubit and uncompute it after preparing all bits. At $n = 16$, this construction uses five clean workspace qubits, 136 counter-controlled rotations, 160 controlled counter-update operations and sixteen local R_Z rotations. These 312 high-level operations exclude the learned layer and do not specify elementary-gate depth. We train using classical shell access and evaluate circuit probabilities through explicit state-vector preparation. The counter construction provides a resource estimate; these experiments do not demonstrate its hardware implementation.

Appendix C: Sixteen-qubit conditional switching construction

1. Graph geometry, cycle reference and symmetry

We order the sixteen edges lexicographically as

$$(01, 02, 03, 04, 05, 06, 12, 13, 14, 15, 16, 23, 24, 25, 26, 34). \tag{C1}$$

The chord set is $\mathcal{C} = \{12, 13, 14, 15, 16, 23, 24, 25, 26, 34\}$. Each chord and its two root edges form a fundamental triangle. Define the binary matrix $B \in \{0,1\}^{10\times 16}$ with one row supported on each such triangle. Its rank is ten over $\mathbb{F}_2$, and the chord columns form the identity matrix. The graph is connected with cycle rank $16 - 7 + 1 = 10$. Six independent vertex switches generate a cut space of size 64. Every cycle row has even incidence at each vertex, so $Bg = 0$ for every switching mask g.

The circuit prepares the reference

$$|\psi_\beta\rangle = \sum_{r\in\{0,1\}^{10}} a_\beta(r)\,|rB\rangle\,, \tag{C2}$$

$$a_\beta(r) = c_\beta^{10-|r|}(-is_\beta)^{|r|}, \tag{C3}$$

where $c_\beta = \cos(t_\beta/2)$, $s_\beta = \sin(t_\beta/2)$, and $t_\beta = (\pi/2)\tanh(1.5\beta)$. Matrix multiplication uses arithmetic modulo two. The physical input is $H^{\otimes 16}\,|\psi_\beta\rangle$. To sample the reference, draw ten independent bits with probability s_β^2 and return rB. The chord bits recover r directly. For a mask $m = uB$ within the cycle code, its sampled amplitude ratio is

$$\frac{a_\beta(r\oplus u)}{a_\beta(r)} = \left[-i\tan\left(\frac{t_\beta}{2}\right)\right]^{|r\oplus u|-|r|}. \tag{C4}$$

Masks outside the code have zero numerator. At $\beta = 0$, only $r = 0$ lies on sampled support; every nonidentity numerator then vanishes. We evaluate ratios only where the sampled denominator is nonzero. The construction does not require enumerating the 2^{16} physical states for reference sampling or amplitude-ratio access.

Cycle orthogonality implies $Z^g\,|\psi_\beta\rangle = |\psi_\beta\rangle$, and the Hadamards give $X^g\,|\phi_\beta\rangle = |\phi_\beta\rangle$. Every IQP gate generated by an X string commutes with these switches. Exact conditional symmetry therefore holds for arbitrary learned IQP phases, although the reported model uses only two shared local angles. Root-versus-chord tying is an intentional parameter restriction. These two classes are not the full edge-orbit decomposition of the seven-vertex graph.

2. Exact quotient evaluation and validation

Let $\ell = Bx$ denote the ten fundamental triangle bits. Each value of ℓ represents exactly sixty-four physical bitstrings. Gauge fixing all root-edge bits to zero gives the chord bits ℓ and an exact energy $E(\ell)$. If ϑ_e denotes the expanded physical phases, then

$$q_{\log}(\ell \mid \beta) = \left|\frac{1}{32}\sum_r a_\beta(r)e^{i\sum_e \vartheta_e(-1)^{(rB)_e}}(-1)^{r\cdot\ell}\right|^2. \tag{C5}$$

The physical distribution is $q(x \mid \beta) = q_{\log}(Bx \mid \beta)/64$. The target has the same orbit multiplicity and satisfies $p_\beta(x) = p_{\log,\beta}(Bx)/64$. Consequently,

$$\frac{1}{2}\sum_{x\in\{0,1\}^{16}} |q(x) - p_\beta(x)| = \frac{1}{2}\sum_{\ell\in\{0,1\}^{10}} |q_{\log}(\ell) - p_{\log,\beta}(\ell)|. \tag{C6}$$

All reported TVD values are therefore exact full physical-distribution values. We also evaluate switching-invariant observables exactly on this quotient. Full balance, $E = 16$, corresponds to one quotient state and sixty-four physical states. This reduction establishes classical tractability for this particular native family; it does not imply a computational advantage.

We validate the cycle preparation and quotient probabilities against direct sixteen-qubit state-vector calculations. We also compare sampled MMD values and gradients with exact Fourier-kernel calculations, including the gradient transformation for tied parameters. These checks assess the circuit and estimator, rather than optimization quality or global model capacity.

3. Data, optimization and model selection

We fit $\beta = 0, 0.4, 0.8, 1.2$ and select at development temperatures $0.18, 0.58, 0.98$. Each fit uses 128 target samples per fitting temperature. The four fixed target pools contain 512 bitstrings in total. Synthetic target draws and exact development distributions come from the finite Gibbs law. The sample count describes training pools, not all target information available to selection. The original reporting temperatures are

$$\begin{aligned}\beta \in \{&0.03, 0.11, 0.27, 0.37, 0.53, 0.67,\\ &0.79, 0.93, 1.09, 1.17, 1.3, 1.4\}.\end{aligned} \tag{C7}$$

Reporting temperatures are excluded from fitting and selection. We treat the experiment as exploratory evidence rather than a fully prospective test of previously unexamined conditions.

At each update, we choose a fitting condition uniformly, sample a batch of 256 targets with replacement, and draw 512 fresh reference samples and 512 nonidentity Fourier masks. We choose either of two Gaussian Hamming-kernel bands uniformly. Their mask bits have Bernoulli probabilities 2/16 and 4/16 before identity exclusion. For a band with bit probability p, each sampled nonidentity mask has weight $[1 - (1 - p)^{16}]/512$. The unbiased finite-sample corrections apply to both target and reference moment squares. We optimize sampled gradients and select checkpoints by exact mean development TVD.

The six candidate configurations cross initial learning rates $0.005, 0.015, 0.04$ with Gaussian initialization widths $0.1, 0.5$. Adam uses $(b_1, b_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and no weight decay. Cosine decay ends at one tenth of the initial learning rate. Six screening fits use 512 updates and checkpoints every 64 updates. The best two candidates each receive two independent 2000-update confirmation fits, with checkpoints every 100 updates. We select the configuration by mean confirmation development TVD, then freeze it before any final reporting evaluation. The selected learning rate is 0.015 with initialization width 0.5. Five independent final fits use 2000 updates. The initial and final checkpoints remain eligible in every stage.

4. Parameter sharing, controls and response diagnostics

An exact-distribution diagnostic at the development temperatures compares one common local angle, root/chord tying with two angles, and seven angles tied to edge orbits under independent exchanges of vertices 1/2, 3/4 and 5/6. The seven groups are $\{01, 02\}$, $\{03, 04\}$, $\{05, 06\}$, $\{12\}$, $\{13, 14, 23, 24\}$, $\{15, 16, 25, 26\}$ and $\{34\}$. Sixteen optimization starts give best observed development TVD values 0.52919, 0.18849 and 0.18322, respectively. These diagnostics fit exact distributions and do not constitute finite-data training results or global optimality certificates. We retain two angles because the observed gain from five additional parameters is 0.00527. The encoder itself uses one fixed temperature map and a fundamental cycle basis, with no extra preparation layers or ancillary qubits.

The encoder-only model is uniform, so its mean balance is zero and its fully balanced probability is 1/1024. Holding each selected model's input at $\beta = 0.6$ yields a sensitivity control, without retraining or selecting a temperature-blind comparator. A separate fixed classical control draws independent edge signs of mean $a_\beta = \sin t_\beta$ and then switches uniformly:

$$q_\beta^{\mathrm{fix}}(x) = \frac{1}{64} \sum_{g \in G} \prod_e \frac{1 + a_\beta (-1)^{x_e \oplus g_e}}{2}. \tag{C8}$$

It has exact symmetry and original-grid mean TVD 0.1134, compared with native TVD 0.1780 ± 0.0013. It limits performance claims but does not change the construction's exact symmetry guarantee.

We compute every observable within each fit, then report the mean and sample standard deviation across five fits. For the Hamiltonian $H = -E$, the per-edge fluctuation statistic is

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2}{16} \left(\langle E^2 \rangle_q - \langle E \rangle_q^2 \right). \tag{C9}$$

It equals equilibrium heat capacity for the Gibbs target; a generated distribution need not satisfy the fluctuation-response relation. At $\beta = 2.4$, the target gives 0.0024 and the native model gives 3.1929 ± 0.0528. Figure 9 displays this discrepancy. We compute the variance within each fit before aggregation, rather than from a mixture of fitted models.

A post-selection diagnostic evaluates the frozen models at 121 points over $\beta \in [0, 2.4]$ using triangle balance and fully balanced probability. We select neither hyperparameters nor checkpoints with these diagnostics. The dense grid includes fitting and development temperatures. Ten further temperatures, $1.5, 1.6, \ldots, 2.4$, provide a separate

post-selection extrapolation summary; we keep it distinct from the original twelve-condition report. Training ends at 1.2, so the displayed interpolation and extrapolation widths are both 1.2. Agreement in the selected order observables does not imply fidelity of all correlations or thermodynamic response.

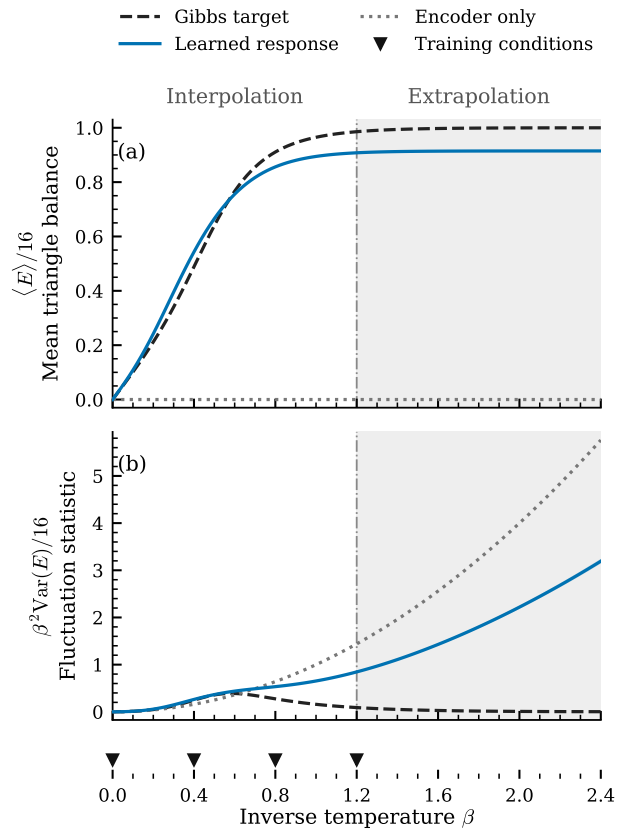


FIG. 9. Supplementary response of the same five frozen fits over $\beta \in [0, 2.4]$. (a) Mean triangle balance, repeated for context. (b) The fluctuation statistic $\beta^2 \,\mathrm{Var}(E)/16$. This statistic equals heat capacity per edge for the equilibrium Gibbs target; the generated distribution need not satisfy the fluctuation-response relation. Styles, sample-standard-deviation bands, fitting markers and shading follow Figure 7. The generated fluctuations remain substantially larger than the target at large inverse temperature. We compute the variance within each fit before aggregation.